Topic: Coefficient of Restitution

The coefficient of restitution (COR), also written with the letter *e*, is used as a measure of the degree to which a collision of two objects is elastic or inelastic. More specifically, the coefficient of restitution is defined as the ratio of the relative velocity of the two objects' separation after the collision to the relative velocity of their approach before the collision:

$$e = \frac{v_{separation}}{v_{approach}}$$

e can also be written as the square root of the ratio of the final kinetic energy of the system to the initial kinetic energy of the system before collision, where velocities are measured relative to the center of mass of the system:

$$e = \sqrt{\frac{KE_{f,rel}}{KE_{0,rel}}}$$

Typically, the coefficient of restitution takes on a value between 0 and 1. A COR of zero represents a completely inelastic collision, where the two objects stick together after collision, and the velocity of separation is zero. A COR of 1 represents a completely elastic collision, where kinetic energy is completely conserved, and the velocity of separation equals the velocity of approach. Most real-world collisions are inelastic and have CORs greater than 0 and less than 1, where some kinetic energy is dissipated in the collision, often through sources such as friction or sound.

Your objective is to design and conduct an experiment related to **Coefficient of Restitution** using at least 2 of the provided materials listed below:

Allowed Materials:

- 1 electronic scale
- 1 piece of cardboard, 1 ft by 1 ft
- 1 ping pong ball
- 1 tennis ball
- 2 rubber bands of different sizes
- 3 sheets of sandpaper with identical grit
- 5 napkins
- 10 cotton rounds
- 30cm masking tape
- 50cm string
- Scissors

You may use any of the above materials in any way to conduct your experiment. In addition to the above materials, each team is allowed to have a timepiece, a linear measuring device, as well as a Class III calculator. No other materials are allowed.

Good luck! :)